Statement of Purpose (SoP)

DSL501: Machine Learning Project

Team Hygieia

August 23, 2025

1. Project Details

- **Project Title:** Synthetic Empathy Across Cultures: Multilingual Detection of Empathy in Hindi and Telugu Mental Health Dialogues
- **Code Repo Link:** `https://github.com/az-raei/DSL501_ML_Project`
- **If Own Idea:** No

2. Problem Statement

India is experiencing a severe mental health crisis. NCRB data indicate that India records the highest absolute number of suicides globally, with a rate of 12.4 per 100,000 people [3]. For over two decades, suicide has been one of the two leading causes of death among young Indians, alongside road accidents [5]. These figures highlight the urgent need for scalable and culturally grounded mental health support.

Empathy is a cornerstone of therapeutic practice. Prior work has demonstrated that empathy can be computationally detected in text-based counseling dialogues in English [4], and that synthetic dialogues can help alleviate data scarcity for empathy modeling [2]. However, most of this work remains limited to English, leaving a gap for Indian languages such as Hindi and Telugu, where expressions of empathy are both linguistically and culturally distinct.

Our project aims to address this gap by extending the EPITOME framework [4] to Hindi and Telugu, leveraging cross-lingual embeddings from XLM-R [1], and generating a modest but culturally validated set of synthetic dialogues.

3. Methodology

Our methodology adapts prior work on computational empathy detection while scaling it down for the scope of a semester project. The main components are:

- **Empathy Framework:** We build on the EPITOME framework [4], which models empathy in counseling dialogues. Empathy is often conceptualized in three dimensions:
  - Cognitive empathy (acknowledging the other's perspective)
  - Affective empathy (emotional resonance)
  - Supportive empathy (comfort or reassurance)

For feasibility in this project, we collapse these into a binary classification task (*empathic* vs. *non-empathic*), while still tracking sub-labels in exploratory analysis.

- **Cross-Lingual Representations:** We integrate XLM-R embeddings [1] to enable transfer learning from English to Hindi and Telugu.
- **Synthetic Dialogue Generation:** Inspired by Lozoya et al. [2], we generate ˜2,000 dialogues per language using large language models, which are then validated by native speakers. Validation focuses on whether generated responses convey empathy, rated by annotators with simple percent-agreement checks.
- **Evaluation:**
  - **Model performance:** Precision, Recall, and F1 score for Hindi and Telugu empathy detection.
  - **Human validation:** Native speaker ratings of generated dialogues for empathy presence.
  - **Cross-lingual consistency:** Checking whether parallel English–Hindi–Telugu dialogues preserve the same empathic intent, reported as percentage agreement across languages.

4. Dataset Details

- **English Baseline:** Reddit counseling dialogues from EPITOME (3,081 pairs) [4].
- **Synthetic Data:** ∼2,000 dialogues each for Hindi and Telugu, generated via LLMs and validated by native speakers.
- **Preprocessing:**
  - Script normalization (Devanagari, Telugu)
  - Handling code-mixing (Hindi-English, Telugu-English)
  - Cultural annotation refinement for empathy categories

5. Required Resources

- **Hardware:** GPU (e.g., NVIDIA T4/V100) for training and fine-tuning.
- **Software:** Python 3.9+, PyTorch, Hugging Face Transformers, IndicNLP toolkit.

6. Novelty of Approach

- First focused extension of synthetic empathy dialogue generation to Hindi and Telugu within a classroom project setting.
- Integration of cognitive and affective empathy categories in multilingual detection.
- Scaled, culturally grounded synthetic data for Indian languages.
- Use of BLEU scores against real EPITOME samples to measure linguistic similarity, ensuring exclusion of low-quality generated samples.

7. Team Composition and Contributions

- **Aalaya:** Responsible for Hindi dialogue generation, annotation, and fine-tuning, mental health conditions information
- **Jagadish:** Responsible for Telugu dialogue generation, annotation, and evaluation metrics. Leads comparative analysis across languages.

8. Expected Outcomes

- Multilingual empathy detection models fine-tuned for Hindi and Telugu.
- A trained empathy detection model evaluated under both binary and multi-class classification.
- ∼4,000 culturally annotated synthetic dialogues.
- Open-source codebase for reproducibility.
- Analytical report comparing empathy across languages, highlighting cultural differences.
- A benchmark comparing original vs. augmented training data using F1, Precision/Recall, and BLEU for generation quality.

9.

References

[1] Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. Unsupervised cross-lingual representation learning at scale. In *Proceedings of ACL*, pages 8440–8451, 2020.

[2] David C Lozoya, Eduardo Hernández Lúa, José A Barajas Perches, Mike Conway, and Simon D'Alfonso. Synthetic empathy: Generating and evaluating artificial psychotherapy dialogues to detect empathy in counseling sessions. In *Proceedings of CLPsych*, pages 157–171, 2025.

[3] NDTV. India has the highest number of suicides globally: Ncrb data, 2024. Accessed: 2025-08-23.

[4] Abhinav Sharma, Adam S Miner, David C Atkins, and Tim Althoff. A computational approach to understanding empathy expressed in text-based mental health support. In *Proceedings of EMNLP*, pages 5263–5276, 2020.

[5] Times of India. For two decades, suicide among top two causes of death for young indians, 2024. Accessed: 2025-08-23.